MILITARY SPECIFICATION

BOMB CLUSTER, INCENDIARY, TH3,

750 - POUND, M36E2 (W/O FUZE, FIN AND WIRE)

1. SCOPE

1.1 This specification covers one type of side opening cluster for use in grouping incendiary bombs for release from aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

E14-23-4081 - Bomb Cluster, Incendiary, TH3, 750-Lb., M36E2 (w/o Fuze, Fin and Wire).
E14-23-4083 - Bomb Cluster, Incendiary, TH3, 750 Lb., M36E2.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing E14-23-4081 or on the subsidiary drawings shall conform to the

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specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Assembly. The cluster shall be assembled (see 6.2) as specified on Drawing E14-23-4081, except that each bomb shall have its safety clamp, if any, removed.

3.3 Leakage. The cluster shall not leak air when subjected to a minimum internal air pressure of 10 inches of mercury gage when tested in accordance with 4.4.4.1.

3.4 Functioning. The cluster shall function so that 95 percent of the bombs shall function upon ground impact when tested and computed in accordance with 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. The cluster shall be free from foreign matter (oil and grease) and damage which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forht in the specificaiton where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Government's responsibility. The Government will perform test specified in 4.3.2.3 and 4.4.4.2. Samples shall be forwarded to the Government test facility designated by the contracting officer.

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 10 clusters shall be manufactured using the same methods, materials, equipment and processes as will be used in regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive test. Each cluster shall be examined in accordance with the classification of defects, the AQL's do not apply. Cluster leakage test shall be 100 percent inspection.

4.3.2.2 For destructive test (functioning). Each cluster shall be tested in accordance with 4.3.2.4.

4.3.2.3 Functioning test (see 6.2). The cluster assembled in accordance with Drawing E14-23-4083 and with the tail fuzes set to function 16 seconds after release from the airplane, shall be dropped from an altitude of 7,000 ± 500 feet traveling at 600 knots indicated airspeed maximum. The fuzes shall function, ignite the detonating cord, and burst the cluster emitting all of the bombs. Should one of the clusters fail to function due to improper fuzing or installation of the cluster, that cluster shall be disregarded and the results computed on the basis of the nine remaining clusters. Data of percent functioning shall be recorded in the following ranges: (See 4.4.4.2 for formula to be used in determining percent functioning.)

(a) 98.1 to 100
(b) 97.1 to 98.0
(c) 95.0 to 97.0

4.3.3 Acceptance/rejection criteria. The preproduction sample (see 6.3) shall meet the examinations and tests specified herein to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168, and shall consist of the clusters produced during one week's production.

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4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be consucted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be based on the percent functioning of the preproduction sample and table I.

Table I. Sampling for functioning test

Percent of bombs functioning in preproduction	Sample size for next 5 lots
98.1 to 100	2 clusters/lot
97.1 to 98.0	3 clusters/lot
95.0 to 97.0	4 clusters/lot

After five production lots have been tested, the percent functioning of these five lots shall be calculated to determine the sample size for the next five production lots, etc. If any one lot falls below 95 percent functioning, that lot shall be rejected, and the next five lots shall have four clusters each dropped. No less than 90 percent of the bombs from satisfactorily dropped clusters shall be recovered. Percent functioning shall be determined in accordance with formula in 4.4.4.2.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. Sample clusters shall be examined in accordance with the classification of defects and MIL-STD-105, except that cluster leakage test shall be 100 percent inspection.

4.4.3.2 For destructive test. Sample clusters shall be tested in accordance with 4.4.4.2.

4.4.3.3 Critical defects. Each cluster in the lot shall be examined for its critical characteristic listed in the classification of defects.

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4.4.3.4 Classification of defects.

(a) Cluster, incendiary bomb, 750lb., M36E2, W/O fuze fin, and wire, assembly (Drawing E14-23-4081).

Categories		Defects
Critical:		
1		Bomb safety plunger exposed
Major:	AQL	0.65 percent defective
101		Component missing or incorrectly assembled
102		Spacer missing
103		Bomb orientation incorrect
104		Detonating cord ends not sealed
105		Workmanship (3.6)

(b) Preparation for delivery (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Strapping loose
102		Identification marking incorrect or illegible
103		Color marking incorrect or illegible

4.4.4 Tests. (See 6.2).

4.4.4.1 Leakage. The cluster shall be subjected to a minimum internal air pressure of 10 inches of mercury gage and observed for pressure drop after 2 minutes. When the pressure drop is more than one thirty-second of an inch of mercury, the cluster is considered to be leaking. (See 6.4). A repaired cluster must pass the test of this paragraph prior to acceptance.

4.4.4.2 Functioning. The cluster assembled in accordance with Drawing E14-23-4083, and with the tail fuzes set to function in 16 seconds after release from the airplane shall be dropped from an altitude of 7,000 ± 500 feet, travelong at 600 knots indicated air speed maximum. The fuze shall function, ignite the detonating cord, and burst the cluster emitting all of the bombs. No less than 90 percent of the bombs from satisfactorily dropped clusters shall be recovered. If 90 percent of the

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bombs are not recovered, additional clusters shall be dropped until 90 percent of the total number are recovered. Percent functioning shall be determined in accordance with the following formula:

$$\text{Percent functioning} = 1 - \frac{\text{number of malfunctioning bombs}}{\text{number of bombs recovered}} \times 100$$

All malfunctioning bombs shall be analyzed for cause of failure (see 6.3).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Not applicable.

5.2 Packing.

5.2.1 Level A, military pack. The cluster shall be packed for shipment in accordance with Drawing D14-23-4073.

5.2.2 Level B, limited military pack. The cluster shall be packed for shipment in accordance with 5.2.1.

5.2.3 Level C, minimum military pack. Not applicable.

5.3 Marking. In addition to any special marking in conjunction with information furnished in contract or order, packing marking shall be in accordance with Drawing D14-23-4073.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.2 Caution. Suppliers are advised to take necessary precautions to protect personnel from injury while assembling and testing the M36E2 cluster.

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6.3 Classification of bombs. When bombs from clusters which have been air dropped are recovered, the following information should be obtained as an aid in determining the cause of failure of each bomb and corrective (if any) taken:

(a) Number of bombs landing flat.
(b) Number of bombs with broken bodies.
(c) *Number of bombs with primers struck but not ignited.
(d) *Number of bombs with primers not struck.
(e) Number of bombs with first fire failure.
(f) Number of bombs with first fire burned but filling not ignited.
(g) Number of bombs with thermate blown out.
(h) Number of bombs exploded (on ground or in air).
(i) Number of bombs not ejected from cluster adapter.

*The angle of impact should be determined for these bombs. When the cause of failure falls in more than one of the above categories only the primary cause shall be considered.

6.4 Repair of leaky clusters. Any cluster found to be leaking may have its leaking point(s) located by immersing the cluster in water containing a suitable wetting agent, subjecteing the cluster to an internal air pressure of 12 ± 2 pounds per square inch gage, and observing for bubbles. The leakage may then be repaired.

6.5 Component specifications. Component specifications for the M36E2 clusters are:

Title	Number
Bomb, Incendiary, 4-lb, TH3, M126	MIL-B-23360
Cord, Detonating	MIL-C-17124
Adapter, Bomb Cluster, 750-lb, M30E2	MIL-A-51372

6.6 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-C-1079, 10 July 1970.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Projcet No. 1325-A330

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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